

Scalable Process for Reduction of Aromatic Carboxylic Acids to Aldehydes via Pinacolborane

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ABSTRACT: The transformation of carboxylic acids to their corresponding aldehydes is one of the most important reactions in both organic syntheses and industrial applications. Our previous research reported an efficient and convenient aldehyde preparative method via controllable reduction of carboxylic acids with pinacolborane (HBpin/DMAP). However, the new method still suffered from some limitations during scaling-up, for example, a large excess of reagents, obvious exotherms, and silica gel column purification. The process development was performed to address these troublesome issues and also to make the new method a practical synthetic protocol. After the detailed exploration of reaction parameters, including the reaction conditions, charging mode, and workup methods, a safe, efficient, and column-free process of reduction of aromatic carboxylic acids to aldehydes via DMAP/HBpin has been successfully established. The newly developed robust process could provide a practical and alternative method for straightforward access of aldehydes from carboxylic acids.

KEYWORDS: process development, reduction of carboxylic acid, aldehyde, pinacolborane (HBpin), scaling-up

1. INTRODUCTION

Aldehydes are extremely important structural motifs as well as crucial starting materials and/or intermediates in the field of synthetic chemistry.¹ Reduction of carboxylic acids (CAs) to aldehydes is one of the most fundamental organic reactions. Over the past century, quite a few methods have been developed for the reduction of CAs and their derivatives to aldehydes.² Compared to the starting CAs, the intrinsically higher reactivity of the resulting aldehydes would pose a challenge to chemoselectivity; i.e., direct reduction of CAs would form aldehydes and/or alcohols. Indeed, most of the reported methods for CAs to aldehydes were indirect approaches or multiple-step reductions.^{2,3} Recently, we have reported a direct and controllable reduction method, which could readily and efficiently convert CAs to their corresponding aldehydes at room temperature.^{4,5} The new method applied HBpin as the reductant together with 4-(piperidin-1-yl)-*N*-trifluoromethylsulfonylpyridin-*N*-ium trifluoromethanesulfonate (Tf-PPDP) as the activator (Scheme 1). Previously, boranes have been extensively applied for the full reduction of CAs and aldehydes to alcohols in both laboratory and industrial scales;^{6–8} thus, our HBpin reduction exhibited the unique value of this controlled reduction of CAs to aldehydes. The newly reported method features easy operation and exhibits high potential for applications in laboratory synthesis.

However, upon scaling up the reaction, some problematic issues were encountered that needed to be addressed. (1) Excess reagents were used. For example, 1.7 equiv of Tf-PPDP, 1.6 equiv of PPDP, and 1.1 equiv of HBpin were necessary for high conversion and yield, and among them, PPDP was quite expensive. (2) Very fast reaction accompanied by a noticeable exotherm was observed during the reaction. The risk of uncontrollable reaction temperature and even occasionally

spraying of the reaction mixture exists. (3) Poor mixing caused by single feeding could lead to the formation of byproducts and would further result in repeatability. (4) The silica gel column purification was not favorable during scaling-up due to concerns about efficiency or effluent disposal. To solve these issues, we have developed a robust process of direct reduction of CAs to aldehydes, featuring safety, controllability, high selectivity, reproducibility, and simple purification suitable for potential scale-up applications.

2. RESULTS AND DISCUSSION

2.1. Initial Reaction Design. To monitor the reaction, 2-naphthoic acid (strong UV response) was chosen as the model substrate for its convenience of in-process control (analyses), including the desired product and byproducts. Since Tf-PPDP (activator) and PPDP (base) showed a marginal advantage over Tf-DMAP (activator) and DMAP (base) in the previous test conditions,⁴ we would like to use the less-expensive DMAP and its derivative Tf-DMAP in the process development.

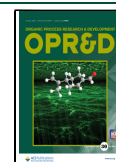
According to the mechanism, the overall reduction involves 4 steps (Scheme 1). Step 1, Tf-DMAP rapidly reacts with CA to form a mixed anhydride (acyl-OTf), releasing the DMAP·TfOH salt. Step 2, the resulting mixed anhydride (acyl-OTf) reacts with 1.0 equiv of DMAP (base) to generate acylpyridinium species (acyl-DMAP, int-1). Step 3, the addition of

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Scheme 1. Pinacolborane Reduction of CAs to Aldehydes and the Proposed Mechanism

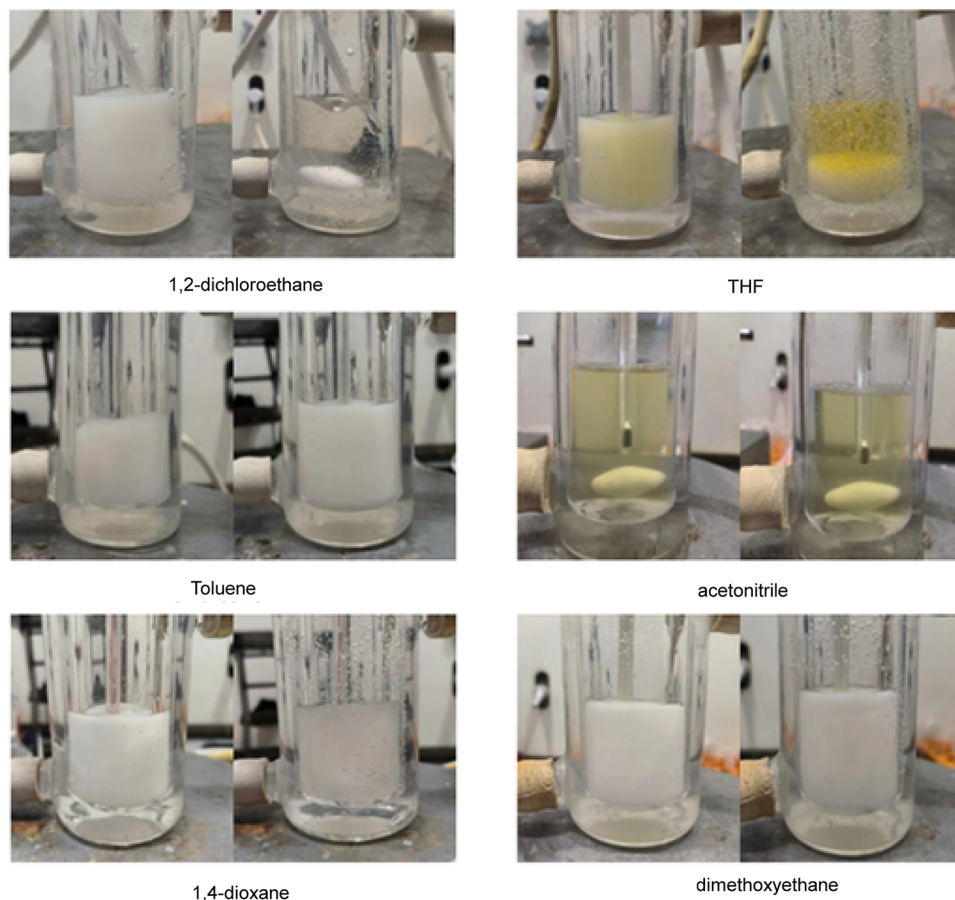
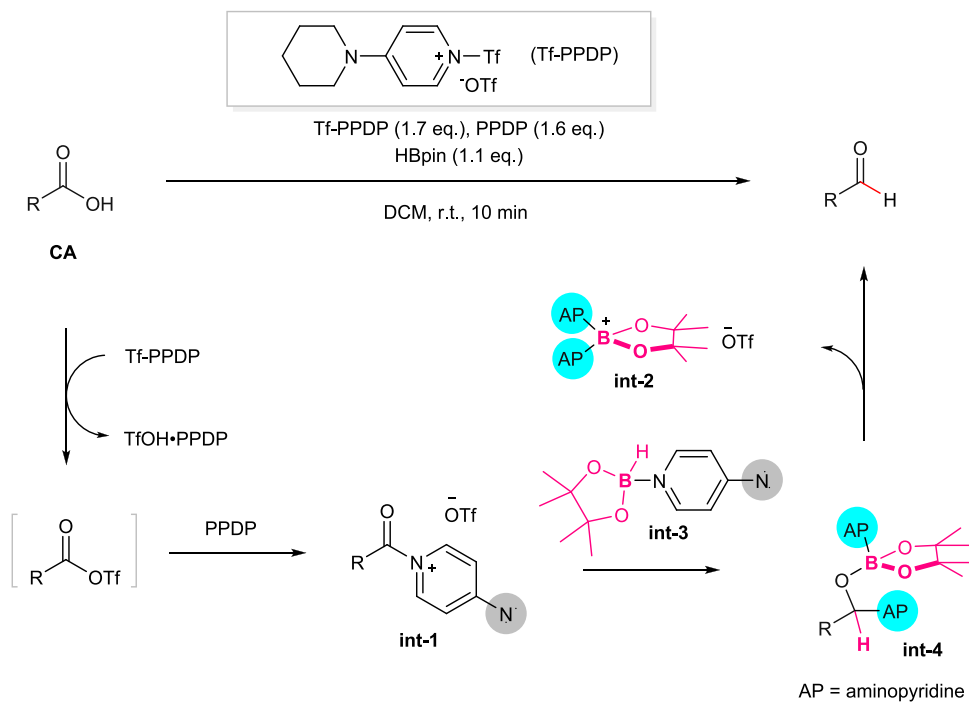
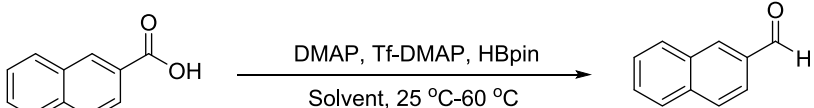


Figure 1. Solubility performance of various solvents in the formation of acyl-DMAP (left: rt; right: 60 °C).

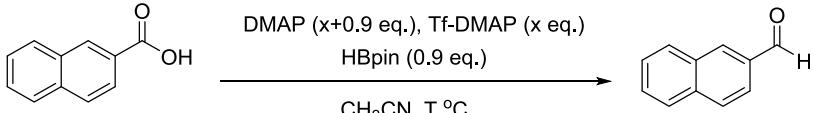
the H–B bond of DMAP·HBpin (**int-3**) to the carbonyl group of acyl-DMAP (**int-1**) gives a C–B adduct (**int-4**). Step 4, the C–B adduct (**int-4**) undergoes an elimination to release a

stable tetra-coordinated B cation complex [(DMAP)₂Bpin]–OTf (**int-2**) and the desired product (aldehyde). Based on the above mechanistic analysis, we assumed that the reaction could

Table 1. Effect of Solvent on the Reduction of Carboxylic Acid to Aldehyde^a


entry	solvent	solubility at 25 °C	solubility at 60 °C	yield/% ^b
1	DCE		yes	62
2	THF			11
3	toluene			64
4	MeCN	yes	yes	84
5	1,4-dioxane		yes	78
6	DME			72

^aReaction conditions: 2-naphthoic acid (3.0 mmol), Tf-DMAP (1.2 equiv), DMAP (2.1 equiv), HBpin (0.9 equiv), solvent (15 mL), 60 °C for 3 h, then 25 °C for 0.5 h. ^bQuantitative analysis by GC with biphenyl as the internal standard.

Table 2. Effect of Tf-DMAP and DMAP Amounts, Reaction Temperature, and Time of Acyl-DMAP Formation on the Reduction of Carboxylic Acid to Aldehyde^a


entry	Tf-DMAP/equiv	DMAP/equiv	temp/°C	time/min	yield/% ^b
1	1.2	1.2 + 0.9	rt	10	77
2	1.2	1.2 + 0.9	rt	180	83
3	1.2	1.2 + 0.9	60	180	84
4	1.2	1.2 + 0.9	60	120	82
5	1.1	1.1 + 0.9	60	180	76
6	1.3	1.3 + 0.9	60	180	69
7	1.4	1.4 + 0.9	60	180	74

^aReaction conditions: 2-naphthoic acid (3.0 mmol), Tf-DMAP (*x* equiv), DMAP (*x* equiv), HBpin (0.9 equiv), CH₃CN (15 mL), *T* (°C), time (h). ^bDetermined by GC analysis with biphenyl as the internal standard.

be simplified into two stages, including the conversion of CA to acyl-DMAP via Tf-DMAP activation and the conversion of acyl-DMAP to aldehyde via HBpin reduction. Based on this assumption, the conversion of CA depends primarily on the transformation of CA to acyl-DMAP, while the selectivity for aldehyde formation is determined by the subsequent reduction with HBpin.

2.2. Solvent. Previous research used dichloromethane (DCM) as the solvent for its better solubility of Tf-PPDP. Unfortunately, Tf-DMAP exhibits limited solubility in DCM. Moreover, the relatively low boiling point of DCM (~40 °C) is not a proper choice for scaling-up due to both the potential exothermic nature of the reaction and its narrow operational temperature range. Thus, various solvents were initially screened to evaluate the solubility of the substrate and reagent, as well as the effects on the reactivity.

According to the mechanism, 1.0 equiv of Tf-DMAP and 1.0 equiv of DMAP were theoretically required for the formation of acyl-DMAP in step 1. The reduction step required 1.0 equiv of DMAP and 1.0 equiv of HBpin, while excess HBpin would result in an over-reduction to yield the alcohol byproduct. Of the above considerations, 1.2 equiv of Tf-DMAP (slightly excess), 2.1 equiv of DMAP (slightly excess), and 0.9 equiv of HBpin (slightly shortage) were employed during the following optimization.

The process was initiatively split into two stages and we abandoned the previous one-pot operation mode. Tf-DMAP, DMAP, and solvent were added into the reaction flask and

stirred at room temperature for 3 min. To the resulting mixture was added 2-naphthoic acid, followed by stirring for an additional 3 min. Solvent test experiments were carried out in 1,2-dichloroethane (DCE), tetrahydrofuran (THF), toluene, acetonitrile (MeCN), 1,4-dioxane, and dimethoxyethane (DME), the results of which are shown in Figure 1. The pictures clearly reflect that only MeCN as a solvent could form a clear solution.

When all slurries were heated to 60 °C, the mixture in DCE turned into a clear solution, while others remained as suspensions. After keeping for 3 h at 60 °C, all reaction mixtures were cooled to room temperature, and then HBpin was added and stirred for 0.5 h. Quantitative analyses via gas chromatography (GC) were carried out with biphenyl as the internal standard. The results are summarized in Table 1. As can be seen, with MeCN as the solvent, the reaction mixture always remained a clear solution, and the preliminary results showed the highest yield (entry 4, Table 1). Therefore, MeCN was chosen for further optimization.

2.3. Charging Amount. According to mechanistic analysis, the conversion of CA was closely related to the ratio of acyl-DMAP and, thus, in turn affected the yield of the desired product. If the mixture contained unreacted CA, it would affect the final yield of both the conversion and the additional consumption of HBpin. After the reaction mixture was quenched, unreacted CA could be observed again and thus make the product separation complicated. With MeCN as the solvent, the formation of acyl-DMAP in the first stage was

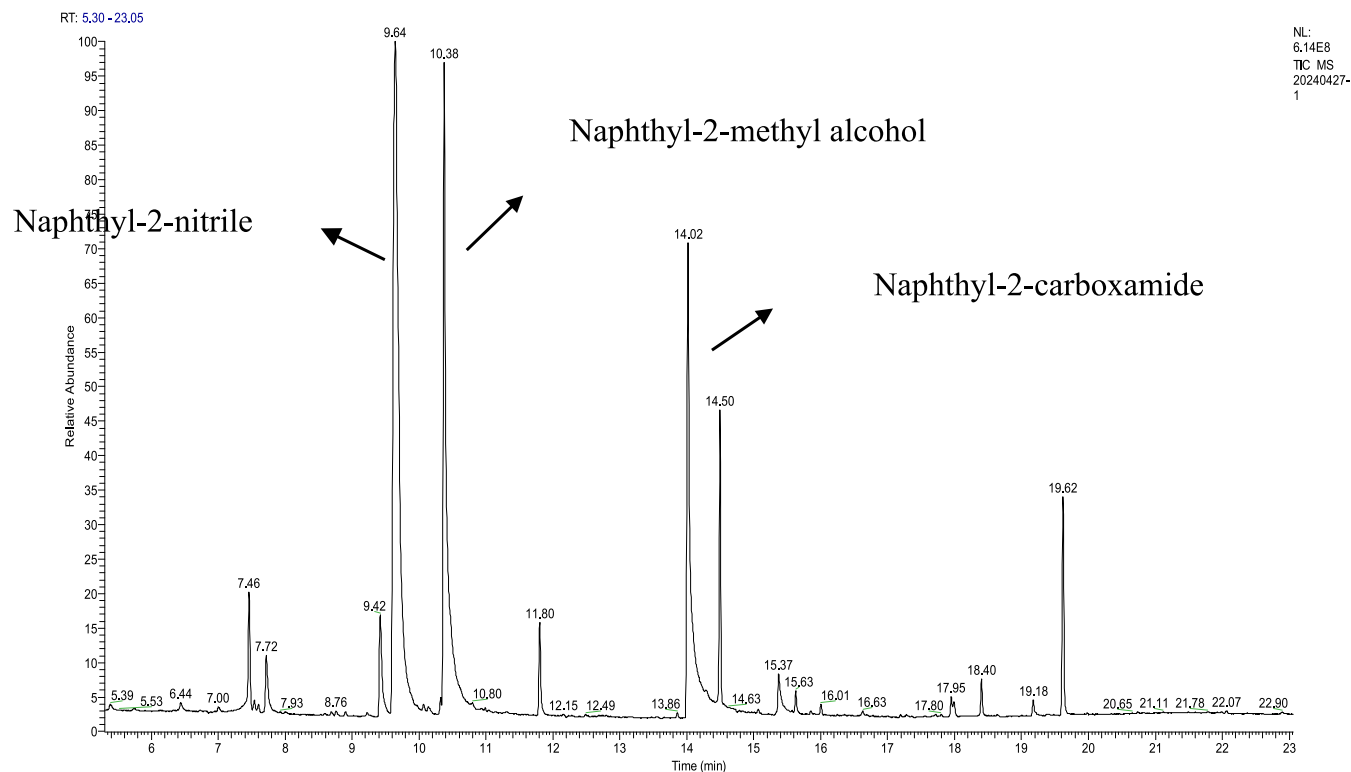


Figure 2. GC–MS spectra of the impurities.

explored for the overall selective reduction. Following the previous procedure, 2-naphthoic acid, 1.2 equiv of Tf-DMAP, and 1.2 equiv of DMAP were mixed at room temperature for 10 min, then another 0.9 equiv of DMAP and 0.9 equiv of HBpin were added. The resulting mixture was stirred for 30 min. A 77% yield of the desired aldehyde was obtained (entry 1, Table 2). While prolonging the first stage to 180 min, the product yield could be improved to 83% (entry 2, Table 2), which indicated that the formation of acyl-DMAP in the first stage was not as fast as previously assumed under ambient conditions. Prolonging the reaction time proved effective; alternatively, increasing the reaction temperature of the first stage could accelerate the reaction (84%, 180 min at 60 °C; entry 3, Table 2). Trying to shorten the reaction time of the first stage, a marginal drop in the yield was observed (entry 4, Table 2). Attempts to improve the yield by adjusting the amounts of Tf-DMAP and DMAP were unsuccessful (entries 4–7).

There was no doubt that something unknown in the reaction mixture resulted in a yield loss. We reckon that a clear reaction profile would help us for further improvement. Column separation was performed for the 3 mmol reaction mixture (entry 3, Table 2). Other than the desired product (394 mg, 84%), 10 mg of over-reduction alcohol byproduct (2%) and 169 mg of unknown mixture were collected. Two major impurities were identified as 2-naphthyl nitrile and naphthylene-2-carboxamide via gas chromatography–mass spectroscopy (GC–MS) analysis for the unknown impurities (Figure 2), which were further confirmed with the authentic samples by thin liquid chromatography (TLC, Figure 3). After scrutinization of the nitrogen source of the impurities, we recognized that the formation of naphthyl-2-nitrile and naphthylene-2-carboxamide could be attributed to the NH_4Cl quench, since the mixture was quenched with aqueous

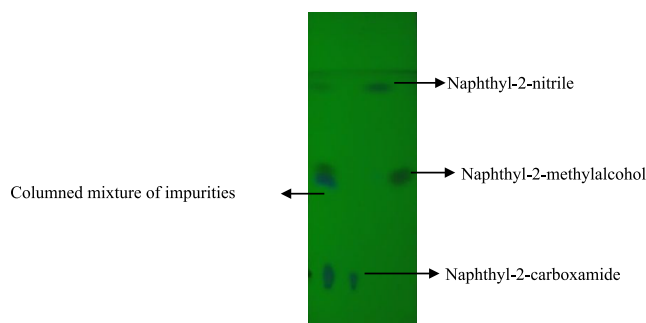
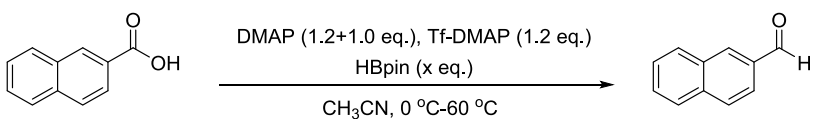


Figure 3. TLC of the impurities.

NH_4Cl solution after the completion of the reaction. It could be explained as follows: the residual intermediate acyl-DMAP was trapped by ammonia to form naphthylene-2-carboxamide, which could undergo dehydration to give 2-naphthyl nitrile. Based on this understanding, we could conclude that the first stage reaction almost reached the full conversion, while 0.9 equiv of HBpin was insufficient for the reduction and thus limited the possibility of higher yield. Further optimization would focus on improving the product yield by reducing the impurities and modifying conditions of the second reduction stage and the workup.

For the second stage of reduction, increasing the amount of HBpin was carefully explored, and the results are summarized in Table 3. As can be seen from Table 3, the yield increased gradually with an increasing amount of HBpin. The highest yield reached up to 91% when 1.0 equiv of HBpin was employed. With a further increase of the amount of HBpin to 1.1 equiv, the yield dropped obviously. The relationship between the yield and the HBpin amount could be well reflected in Figure 4.

Table 3. Effect of HBpin Amount on the Reduction of Carboxylic Acid to Aldehyde^a

		
entry	HBpin/equiv	yield/% ^b
1	0.9	82
2	0.95	87
3	0.98	90
4	1.0	91
5	1.05	90
6	1.1	85

^aReaction conditions: 2-naphthoic acid (3.0 mmol), Tf-DMAP (1.2 equiv), DMAP (2.2 equiv), HBpin (*x* equiv), CH₃CN (15 mL), 0–60 °C, 3.5 h. ^bDetermined by GC analysis with biphenyl as the internal standard.

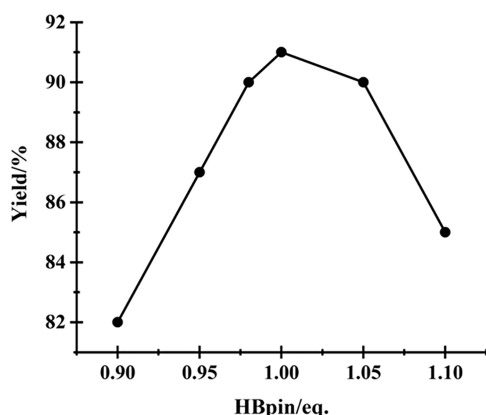


Figure 4. Relationship between the yield of aldehyde and the HBpin amount.

2.4. Charging Mode. After optimizing the amount of HBpin, the charging mode was carefully considered due to the risk of the exotherm. The results are summarized in Table 4. Premixing of DMAP and HBpin accordingly formed a DMAP·HBpin complex, which might effectively activate HBpin for acyl-DMAP reduction. These experiments showed that (1) single DMAP charging was unfavorable to the reaction (entry 1

vs entry 2; entry 3 vs entry 4, Table 4); (2) addition of DMAP·HBpin/MeCN solution to the acyl-DMAP/MeCN solution was essential and brought the operational convenience. On the other hand, reverse addition of acyl-DMAP/MeCN to DMAP·HBpin/MeCN was demonstrated to be an unsuitable charging mode (entry 1 vs entry 6, Table 4). TLC monitoring of the reaction indicated that quite a lot of alcohol byproduct was generated and unreacted carboxylic acid remained (Figure 5).

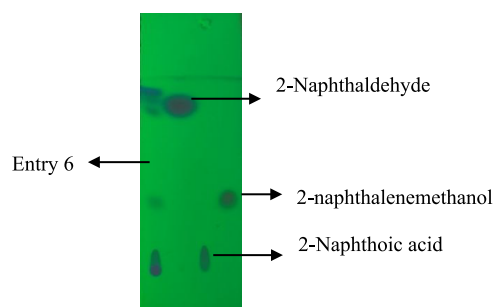


Figure 5. TLC of entry 6.

This result was not beyond the expectation that the generated active aldehyde would quickly further reduce under the condition of high content of HBpin in the pot-bottom; (3)

Table 4. Effect of the Charging Method on the Reduction of Carboxylic Acids to Aldehydes^a

Reaction scheme: 2-naphthoic acid reacts with DMAP, Tf-DMAP, and HBpin in CH₃CN at 25 °C–60 °C to produce 2-naphthaldehyde.

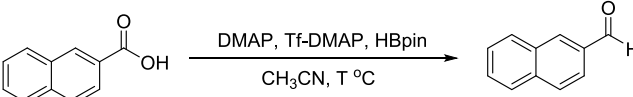
entry	charging mode	temperature increase/ °C	yield/% ^b
1	Tf-DMAP; DMAP (1.2 equiv) as the pot-bottom, rt, addition of acid; addition of DMAP (0.9 equiv) and HBpin		82
2	Tf-DMAP; DMAP (2.1 equiv) as the pot-bottom, rt, addition of acid; addition of HBpin		71
3	acid; DMAP (1.2 equiv) as the pot-bottom, rt, addition of Tf-DMAP; addition of DMAP (0.9 equiv) and HBpin		63
4	acid; DMAP (2.1 equiv) as the pot-bottom, rt, addition of Tf-DMAP; addition of HBpin		58
5	acid as the pot-bottom, rt, addition of Tf-DMAP and DMAP (1.2 equiv); addition of DMAP (0.9 equiv) and HBpin		72
6	reverse addition of the first stage mixture to the solution of DMAP (0.9 equiv) and HBpin (0.9 equiv)		trace
7	Tf-DMAP; DMAP (1.2 equiv) as the pot-bottom, rt, addition of acid; addition of DMAP (1.0 equiv) and HBpin		91
8 ^c	Tf-DMAP; DMAP (1.2 equiv) as the pot-bottom, rt, addition of acid; addition of DMAP (1.0 equiv) and HBpin	16	92 ^d
9 ^c	acid; DMAP (1.2 equiv) as the pot-bottom, rt, addition of Tf-DMAP; addition of DMAP (1.0 equiv) and HBpin	4	90 ^d
10 ^c	acid; DMAP (1.2 equiv) as the pot-bottom, 60 °C, addition of Tf-DMAP; addition of DMAP (1.0 equiv) and HBpin	0	92 ^d

^aReaction conditions: 2-naphthoic acid (3.0 mmol), Tf-DMAP (1.2 equiv), DMAP (*x* equiv), HBpin (*x* equiv), CH₃CN (15 mL), 25–60 °C, 3.5 h. ^bDetermined by GC with biphenyl as the internal standard. ^cReaction conditions: 2-naphthoic acid (30.0 mmol), Tf-DMAP (1.2 equiv), DMAP (2.2 equiv), HBpin (1.0 equiv), CH₃CN (120 mL), 0–T °C, 3.5 h. ^dIsolated yield.

with Tf-DMAP and 1.2 equiv of DMAP and MeCN as the pot-bottom, addition of the starting carboxylic acid/MeCN solution in 5 min at room temperature led to a relatively high yield. However, this charging method exhibited an obvious exotherm, a 16 °C temperature increase was observed as the reaction scaled up to 30 mmol (entry 8, Table 4). This exotherm could be ascribed to the high concentration of Tf-DMAP in the pot-bottom. Considering that the highly active Tf-DMAP resulted in a temperature increase, charging mode was adjusted as follows: charging the carboxylic acid, DMAP, and MeCN as the pot-bottom, then addition of Tf-DMAP/MeCN solution (0.4 g/mL) after 30 min at room temperature. This modification significantly alleviated the exotherm, and only 4 °C temperature increase was observed (entry 9, Table 4). Because the mixture in the first stage was heated to 60 °C to promote the full conversion of acyl-DMAP, the charging mode was then modified further as direct addition of Tf-DMAP/MeCN to the 60 °C pot-bottom mixture of carboxylic acid and DMAP/MeCN solution. This variation simplified the operation, controlled the temperature increase, and afforded a high yield (entry 10, Table 4); more importantly, charging at a higher temperature led to no temperature increase.

The above modification of the charging mode effectively addressed the safety concern and primarily maintained the yield. However, the quick reduction of the previous procedure of the second stage with a single-charging HBpin would also be inappropriate for the potential scaling-up. The addition mode was naturally cited in the optimization. Meanwhile, the reduction temperature was parallelly explored with the addition of HBpin solution, and the results are listed in Table 5. High-reduction temperature was unbeneficial to the yield of the desired aldehyde, accompanied by quite a lot of alcohol byproducts (entries 1 and 2, Table 5). As the reduction was performed at a lower temperature, the yield was obviously increased (entries 3 and 4, Table 5). A further decrease of the reaction temperature to −10 °C gave a slurry mixture, thus leading to a mixing issue, although an acceptable reduction

Table 5. Effect of Reduction Temperature, Drop Time, and Reaction Time on the Reduction of Carboxylic Acid to Aldehyde^a

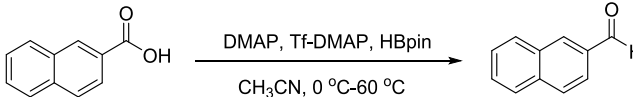
				
entry	addition temperature/°C	addition period/min	stirring period/min	yield/% ^b
1	60	15	direct quenching after addition	28
2	rt	15	direct quenching after addition	40
3	5	15	30	62
4	0	15	30	61
5	−10	15	30	59
6	0	2	30	91
7	0	2	10	87
8	0	2	45	90
9	0	2	60	90

^aReaction conditions: 2-naphthoic acid (3.0 mmol), Tf-DMAP (1.2 equiv), DMAP (2.2 equiv), HBpin (1.0 equiv), CH₃CN (15 mL), T °C, T h. ^bDetermined by GC analysis with biphenyl as the internal standard.

yield was observed (entry 5, Table 5). 0 °C was the optimal temperature for the following optimization. It is easy to understand that the later high concentration of aldehyde would compete with the low concentration of the acyl-DMAP as the slow addition progressed in the reduction. A quick addition of HBpin solution is therefore essential (entry 6 vs entry 4, Table 5). Then, stirring of the reaction mixture was done after completing the addition. The results showed that stirring at 0 °C for 30 min is suitable (entries 6–9, Table 5). The optimal conditions for the second stage were the quick addition of HBpin/MeCN over 2 min, followed by stirring for 30–60 min at 0 °C.

2.5. Quenching and Extraction Conditions. A satisfied yield was obtained after the optimization of reaction conditions, including the molar ratio, charging mode, and reaction temperature. Optimization of the workup procedure is also important for the final separation. Generally, the reaction mixture was adjusted to a weak acidic environment, which could make DMAP stay in the aqueous phase in a salt form (DMAP·HCl). Then, the desired product was extracted with a solvent. The results of quenching and isolation are summarized in Table 6. The initial quench with NH₄Cl could efficiently

Table 6. Effect of the Quenching Method and Extraction Conditions on the Yield of the Reaction^a

				
entry	quenching reagent	extraction solvent	extraction temp/°C	yield/% ^b
1	satd. NH ₄ Cl	DCM	rt	82
2 ^c	aq citric acid	DCM	rt	72
3	aq citric acid	DCM	rt	76
4	aq citric acid	toluene	rt	82
5	ice water	toluene	60	85
6	ice water	Me-THF	60	77
7 ^d	ice water	<i>n</i> -heptane	60	82
8 ^d	aq citric acid	toluene	60	93
9 ^d	aq citric acid	toluene/ <i>n</i> -heptane (1:5)	60	91
10 ^d	aq citric acid	toluene/ <i>n</i> -heptane (1:2)	60	90

^aReaction conditions: entries 1–10: 2-naphthoic acid (3.0 mmol), Tf-DMAP (1.2 equiv), DMAP (2.2 equiv), HBpin (1.0 equiv), CH₃CN (x mL), 0–60 °C, 3.5 h. ^bGC quantitative yield with biphenyl as the internal standard. ^cThe reaction mixture was reverse-added to aqueous citric acid solution, and extraction was carried out three times. ^dCH₃CN was removed *in vacuo* after quenching.

transfer DMAP into the aqueous phase. However, naphthyl-2-nitrile and naphthyl-2-carboxamide were generated during the quenching procedure as previously mentioned (Figures 2 and 3), and the yield (GC quantitative analysis) was not as good as expected (entry 1, Table 6). Later, an aqueous citric acid solution was employed to quench the reaction mixture. Three extractions either with DCM or toluene could not provide the satisfied yield (entries 2–4, Table 6). It was assumed that cold water could avoid the potential byproduct during the quenching; however, similar phenomena were observed with ice water as the quenching reagent (entries 5–7, Table 6). The unsatisfied yield using aqueous citric acid or ice water for the quenching step could be attributed to the fact that the high

Table 7. Salting-Out Effect and Effect of Extraction Conditions on the Yield of the Reaction^a

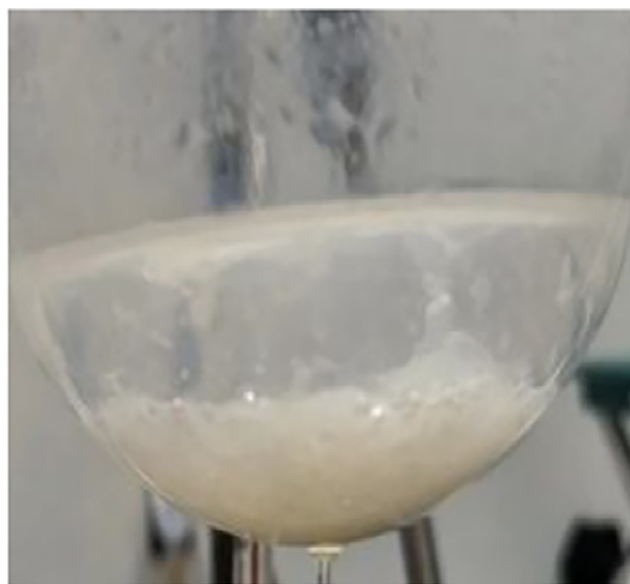
entry	salt	extraction solvent	E1/g (%)	E2/g (%)	E3/g (%)	yield/% ^b
1	NaCl	<i>n</i> -heptane	1.03 g (66.0) ^c	0.30 g (19.2)	0.08 g (5.1)	90.3
2	NaCl	<i>n</i> -heptane:toluene = 5:1	1.12 g (71.8)	0.26 g (16.4)	0.04 g (2.7)	90.9
3	Na ₂ SO ₄	<i>n</i> -heptane	1.02 g (65.8)	0.29 g (18.6)	0.08 g (5.1)	89.5
4	MgSO ₄	<i>n</i> -heptane	1.03 g (67.2)	0.31 g (19.8)	0.07 g (4.4)	91.4
5	CaCl ₂	<i>n</i> -heptane	0.91 g (57.8)	0.43 g (27.3)	0.02 g (1.3)	86.4

^aReaction conditions: 2-naphthoic acid (10.0 mmol), Tf-DMAP (1.2 equiv), DMAP (2.2 equiv), HBpin (1.0 equiv), CH₃CN (*x* mL), 0–60 °C, 3.5 h. The reaction mixture was quenched with aqueous inorganic salt solution. CH₃CN was removed *in vacuo* after quenching. Extraction was done three times at 60 °C. ^bIsolated yield. ^cData in parentheses are the isolated yield for every extraction. E1 = first extraction, E2 = second extraction, and E3 = third extraction.

ratio of MeCN in the aqueous mixture was detrimental to the extraction efficiency. To verify this speculation, most MeCN in the aqueous citric-acid-quenched mixture was removed *in vacuo* before extraction. Indeed, the following extraction efficiency was significantly improved (entries 8–10, Table 6). Meanwhile, four solvents were tested for the extraction. 2-Methyltetrahydrofuran (2-Me-THF) showed high extraction efficiency, but the extracted phase contained both the alcohol byproduct and quite a lot of DMAP. DCM exhibited a relatively low efficiency, yet contained both alcohol and DMAP. Toluene extraction showed a rather clean profile, with most being the product, together with some alcohol byproduct and a slight amount of DMAP. Comparing the polarity of the above tested solvents, we assumed that *n*-heptane could balance the solvent polarity and solubility, thus a mixed solvent of toluene and *n*-heptane was utilized. The mixed solvent system did work well and provided an almost clean product solution. When the extraction was performed at 60 °C, the extraction efficiency was significantly improved once again. Considering that the mixed solvent system would potentially lead to a high effluent, single *n*-heptane was tested for extraction at 60 °C. It was found that 82% of clean product was obtained with a combination of extracts three times (entry 7, Table 6, detailed TLC diagrams of extraction, also see the Supporting Information). Thus, a workup procedure of quench and column-free purification was primarily established. The optimized workup process greatly simplified the purification.

2.6. Salting-Out Effect and Isolation Yield. Although *n*-heptane worked well for extraction at 60 °C, a gap of 10 percentage was observed after three times of extraction. As a common knowledge, some inorganic salts could make an obvious salting-out effect on improving the extraction efficiency. With the precise isolation yield, a reduction reaction was carried out in a 10 mmol scale with the optimal conditions, and then various salts were tested (Table 7). To ensure the accuracy, a single *n*-heptane and a mixed solvent system were paralleled. Under the same conditions, *n*-heptane and the mixed solvent system showed negligible differences with NaCl aqueous solution (entry 1 vs entry 2, Table 7). For all tested salts, MgSO₄ provided the highest yield (up to 91.4%) and clean product (entry 4, Table 7). However, CaCl₂ did not perform well either in phase-out (Figure 6) or in terms of yield (entry 5, Table 7).

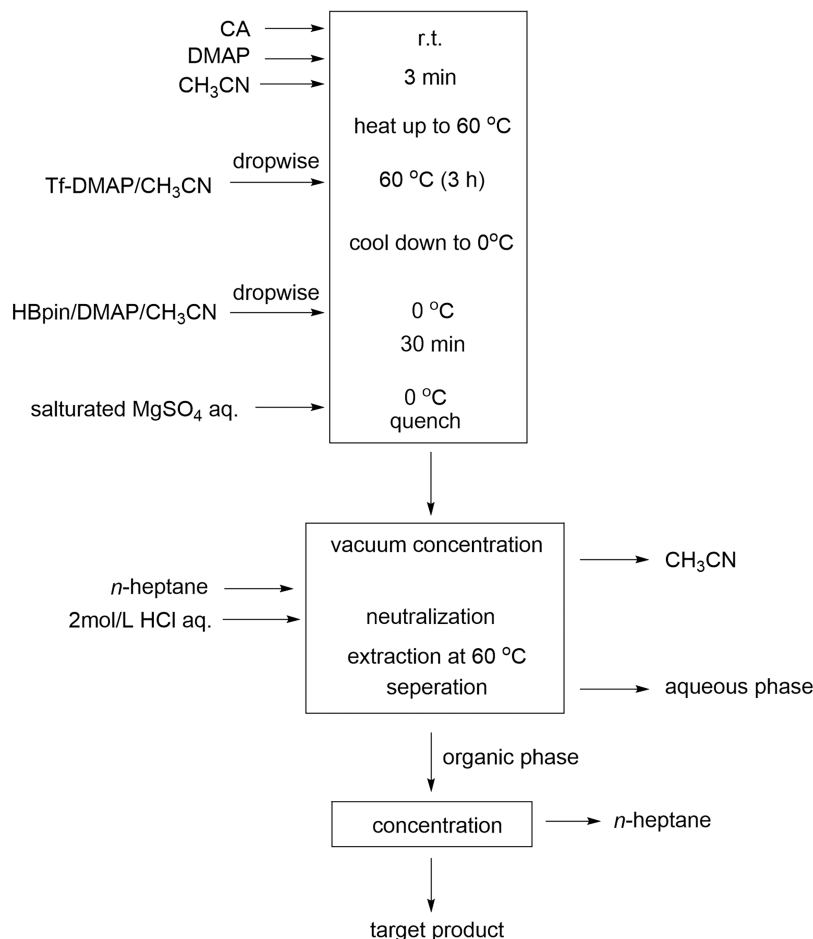
2.7. Optimal Process Definition. Up to now, the optimal condition for the HBpin reduction of carboxylic acid to aldehyde could be concluded as follows. To the MeCN

Figure 6. Second extraction of aq CaCl₂ solution.

solution of carboxylic acid (10 mmol) and DMAP (1.2 equiv) at 60 °C, the solution of Tf-DMAP (1.2 equiv)/MeCN was dropwise added within 30 min. Then, the reaction mixture was stirred for an additional 3 h. The resulting mixture was cooled to 0 °C under a nitrogen atmosphere. To the cooled solution was added a solution of DMAP (1.0 equiv) and HBpin (1.0 equiv)/MeCN in 2 min. The mixture was stirred for 30 min at 0 °C, then quenched with aqueous MgSO₄ solution. MeCN was removed *in vacuo*. *n*-Heptane was added, and the pH was adjusted to neutral using 2 mol/L HCl solution. The resulting mixture was heated to 60 °C, then extracted with *n*-heptane three times. The solvent of the combined *n*-heptane phases was removed *in vacuo* to afford the desired aldehyde product (Chart 1).

2.8. Gradual Scaling-Up. With the optimized process in hand, scaling-up experiments were carried out to verify the reliability of the newly developed process. Again, naphthyl-2-carboxylic acid was chosen as the substrate. The reduction experiments were gradually scaled up from 516 mg (3 mmol) to 25.8 g (150 mmol). All reactions worked smoothly and afforded the desired product in high yield (91–94%) (entries 1–4, Table 8). It is pleasing that the process is also suitable for other aromatic acids. For example, the reduction of an industrially relevant intermediate for pharmaceuticals and

Chart 1. Flowchart of HBpin Reduction of Aromatic Carboxylic Acid to Aldehyde

Table 8. Scale-Up Reactions for Aromatic Carboxylic Acid Reduction to Aldehyde^a

$\text{Aryl}-\text{C}(=\text{O})\text{OH} \xrightarrow[\text{CH}_3\text{CN}, 0\text{ }^\circ\text{C}-60\text{ }^\circ\text{C}]{\text{DMAP, Tf-DMAP, HBpin}} \text{Aryl}-\text{C}(=\text{O})\text{H}$			
Entry	Product	Scale (mmol)	Yield/% ^b
1	 1a	3	0.4 g (91%)
2		10	1.5 g (94%)
3		30	4.3 g (92%)
4		150	21.5 g (92%)
5	 2b	5	0.7 g (87%)
6		30	4.4 g (92%)
7		150	21.4 g (89%)

^aReaction conditions: carboxylic acid (*x* mmol), Tf-DMAP (1.2 equiv), DMAP (2.2 equiv), HBpin (1.0 equiv), CH₃CN (*x* mL), 0–60 °C, 3.5 h.

^bIsolated yield.

agrochemicals such as 1-methyl-3-difluoromethyl-pyrazol-4-carboxylic acid could be conveniently converted to its corresponding aldehyde in excellent yield in every scale (entries 5–8).

3. CONCLUSION

Based on the innovative chemistry of Tf-DMAP-mediated HBpin reduction of carboxylic acid, we have successfully developed a robust reduction process of aromatic carboxylic

acid to its corresponding aldehyde. The new process featured the advantages of safety, cost-effectiveness, “greenness”, being column-free, and high efficiency. The new process employed less expensive and more readily available DMAP to replace the relatively expensive PPDP (cost). The current work intends for rapid conversion of carboxylic acid to aldehyde on a subkilogram scale. Safety and toxicity concerns have not been encountered, and the reagent could be operated safely following the standard procedure. For industrial-scale

production, additional considerations such as large-scale handling and waste management would be required. MeCN was used as the only solvent, which avoided the use of halogenated solvents (greener). Addition of Tf-DMAP/MeCN solution at 60 °C could ingeniously eliminate the potential exotherm (safety concern). Quick addition of DMAP and HBpin/MeCN solution but not a single shot at 0 °C significantly improved the selectivity (controllable). Finally, hot extraction with *n*-heptane assisted by MgSO₄ efficiently provided a high yield of clean product and eliminated tedious column purification. The process was smoothly demonstrated in a 150 mmol scale.

4. EXPERIMENTAL SECTION

4.1. General. All ¹H and ¹³C NMR spectra were recorded on a Bruker DRX 500 MHz spectrometer in deuterium solvent with tetramethylsilane (TMS) as the internal standard. NMR multiplicities are abbreviated as follows: s = singlet, d = doublet, t = triplet, q = quartet, sept = septet, m = multiplet, and br = broad signal. Chemical shifts are given in parts per million and are referenced to TMS (¹H and ¹³C). GC analysis was performed on an Agilent 7890A instrument with a flame ionization detector (FID). An HP-5 capillary column (30 m, 0.32 mm i.d., 0.25 μm film thickness) was used. GC–MS analysis was performed on a Thermo Trace ISQ instrument with a xx detector. A TG SMS capillary column (60 m, 0.25 mm i.d., and 0.25 μm film thickness) was used. All chemicals were commercially available and used as received.

4.2. Synthesis of Tf-DMAP. To a 2000 mL three-necked jacketed flask with a mechanic stirrer were added 122.2 g of DMAP (1.0 mol) and 1.6 L of dichloromethane. The mixture was stirred until the solid dissolved completely. The temperature of the mixture was cooled to 0 °C. 352.7 g of Tf₂O (352.7 g, 1.2 mol, 1 equiv) was added slowly into the flask over 1.5 h. Then, the circulating water bath was removed. The temperature of the mixture was gradually raised to room temperature and stirred for additional 4 h. After filtration, elution with dichloromethane, and drying under vacuum, 375.7 g of Tf-DMAP was obtained as a white solid (yield: 93%). ¹H NMR (500 MHz, CD₃CN) δ 8.29 (d, J = 8.0 Hz, 2H), 7.16 (d, J = 8.0 Hz, 2H), 3.41 (s, 6H) ppm.

4.3. Typical Procedure for the Reduction of Aromatic Carboxylic Acid to Aldehyde. To a 1000 mL four-necked jacketed flask with a mechanic stirrer were added 25.8 g of 2-naphthoic acid (0.15 mol), 22.0 g of DMAP (0.18 mol, 1.2 equiv), and 120 mL of acetonitrile. The mixture was stirred for 3 min and heated to 60 °C. A solution of Tf-DMAP (72.8 g, 0.18 mol, 1.2 equiv) in 180 mL of acetonitrile was prepared separately and added dropwise to the jacketed flask over 1 h. The mixture was stirred at 60 °C for 3 h, then cooled to room temperature and subsequently to 0 °C using a circulating cold bath. Meanwhile, to another 250 mL round-bottom flask were added 18.3 g of DMAP (0.15 mol, 1.0 equiv), 100 mL of acetonitrile, and 19.2 g of HBpin (0.15 mol, 1.0 equiv), affording a clear solution after stirring for 3 min. The solution of HBpin-DMAP was added dropwise to the cooled reaction mixture (0 °C) over 40 min. After being stirred for 30 min, the mixture was quenched with 80 mL of saturated aq MgSO₄. The solvent was removed under reduced pressure. 300 mL of *n*-heptane was added to the aqueous phase, and the biphasic mixture was heated to 60 °C. The aqueous phase was adjusted to neutral pH with aq HCl (2 mol/L). Then, the mixture was stirred at 60 °C for 1 h, and the aqueous phase was separated.

The extraction with *n*-heptane was repeated twice more, and all of the organic phase was collected. After drying with Na₂SO₄, filtration, and concentration, 21.5 g of 2-naphthaldehyde was obtained as a white solid (yield: 92%). ¹H NMR (500 MHz, CDCl₃) δ 10.16 (s, 1H), 8.33 (s, 1H), 8.03–7.87 (m, 4H), 7.68–7.63 (m, 1H), 7.62–7.57 (m, 1H) ppm.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.oprd.5c00344>.

NMR spectra of Tf-DMAP, 2-naphthaldehyde, and 3-(difluoromethyl)-1-methyl-pyrazole-4-carbaldehyde (PDF)

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Notes

The authors declare no competing financial interest.

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